DUDALA GANESH

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EXPERIENCE SUMMARY

- 1+ year of professional experience in Data Science, Machine Learning, Generative AI and Agentic AI systems, implementing and deploying AI-driven applications.
- Completed 0.2 year Machine Learning Internship at DRDO (Defence Research and Development Organization), implementing ML solutions for drone surveillance analytics, anomaly detection and terrain classification using TensorFlow and PyTorch.
- Developed enterprise AI assistants and chatbots using NLP, Deep Learning and LLMs such as GPT-4 and Azure OpenAI.
- Domain experience across Sales, Human Resources (HR) and Enterprise AI Assistant solutions.
- Designed and implemented Agentic AI architectures and multi-agent systems using LangChain, CrewAI and LangGraph for autonomous reasoning, task orchestration and planning- and memory-driven agent workflows.
- Implemented Retrieval Augmented Generation (RAG) and Agentic RAG architectures integrating Pinecone, FAISS and Neo4j for semantic search and knowledge retrieval.
- Built multimodal AI agents capable of processing text, image and structured enterprise data for enhanced contextual understanding.
- Developed scalable AI APIs and backend services using FastAPI to support production deployment and concurrent AI request handling, integrating external tools and REST APIs to augment agent capabilities.
- Implemented machine learning models such as Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, K-Means and XGBoost for predictive analytics.
- Developed deep learning solutions using Neural Networks, Transformers and STGCN, including traffic flow prediction models using the PEMS dataset.
- Worked with cloud platforms such as AWS (Bedrock, CodeCommit) and Microsoft Azure, including Docker-based containerized deployment, for integrating AI/ML and GenAI applications.
- Applied responsible AI practices by monitoring model performance, validating AI outputs and continuously testing agent pipelines for accuracy, reliability and safety.

EDUCATION

Bachelor of Technology (AIML) — Sreyas Institute of Engineering & Technology, Hyderabad (2022 – 2025)

TECHNICAL SKILLS

Programming Languages	Python, SQL, TypeScript
Machine Learning	Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Means Clustering, XGBoost
Deep Learning & AI	Neural Networks, Transformers, Deep Learning, Spatio-Temporal Graph Convolutional Networks (STGCN)
NLP & Generative AI	Large Language Models (LLMs), GPT-4, Prompt Engineering, Retrieval Augmented Generation (RAG), Agentic RAG, NL-to-SQL, Text Processing
Agentic AI & AI Frameworks	LangChain, CrewAI, LangGraph, Multi-Agent Systems, Agent Engineering, AgentOps, Agent Planning & Memory Modules, AI Agent Workflows, n8n
AI/ML Frameworks	TensorFlow, PyTorch, Scikit-learn, Statsmodels
Data Processing & Analysis	Pandas, NumPy, OpenPyXL, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA)
Backend & APIs	FastAPI, Uvicorn, REST APIs, Pydantic, Tool & API Integration for Agents
Frontend	React.js, TypeScript, Vite, TanStack Query, Chart.js, Tailwind CSS
Databases & Data Processing	DuckDB, SQLite, Pandas, ETL Pipelines
Vector Databases & Knowledge Retrieval	Pinecone, FAISS, Neo4j, OpenAI Embeddings, Semantic Search
Cloud & DevOps	AWS (Bedrock, CodeCommit), Microsoft Azure, Azure OpenAI Services, Azure Container Apps, Azure File Share, Azure Active Directory (MSAL), Docker, Nginx
Responsible AI	Model Output Validation, Performance Monitoring, Safety & Reliability Testing for LLM Systems

Development Tools	Git, VS Code, Postman, Jupyter Notebook, Matplotlib

PROFESSIONAL EXPERIENCE

R Systems International
Associate Data Scientist (Agentic AI Engineer) May 2025 – Current

Project #1: Zyra 2.0 – AI-Powered HR Recruitment & Attrition Intelligence Platform
Domain: HR AI Assistant Services / Workforce Intelligence
Duration: Feb 2026 – Current
Technology: Python, TypeScript, SQL, LangChain, Azure OpenAI, XGBoost, Scikit-learn, Statsmodels, FastAPI, React.js, DuckDB, Docker, Azure

Description: Designed and developed an enterprise-grade AI-powered HR analytics platform delivering recruitment insights, workforce intelligence, predictive analytics and natural language querying over HR data. Combines Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG) and interactive dashboards to automate hiring analytics, predict employee attrition, forecast recruitment demand and enable conversational analytics through an AI chatbot. Deployed on Microsoft Azure using containerized services and integrated with Azure OpenAI for intelligent business insights.

Responsibilities:
- Designed and developed an end-to-end AI-powered HR analytics platform for recruitment, attrition analysis and workforce intelligence.
- Built a LangChain-based RAG pipeline with Azure OpenAI to enable natural language querying over HR datasets, incorporating contextual memory for multi-turn agent reasoning.
- Developed an NL-to-SQL chatbot capable of answering business questions using conversational AI and contextual memory.
- Implemented Machine Learning models using XGBoost for early employee attrition prediction and recruiter performance evaluation.
- Developed time-series forecasting models to predict hiring demand and fulfillment trends.
- Created RESTful APIs using FastAPI to serve AI models, dashboards, chatbot services and analytics, integrating external tools and data sources to augment agent capabilities.
- Designed and optimized ETL pipelines for processing structured and semi-structured HR data from Excel into DuckDB.
- Integrated Azure OpenAI to generate AI-driven insights, summaries and analytical reports for HR users.
- Built scalable RAG workflows using embeddings, semantic search, vector retrieval and prompt engineering techniques.
- Implemented Azure File Share synchronization with automated data versioning, retraining and cache refresh mechanisms.
- Developed role-based authentication and authorization using Azure Active Directory (Azure AD/MSAL).
- Containerized frontend and backend applications using Docker and deployed them on Azure Container Apps.
- Optimized backend performance through caching, query optimization and asynchronous API processing.
- Collaborated with cross-functional teams to gather business requirements and deliver production-ready AI solutions.
- Monitored model performance, validated AI outputs and continuously improved system accuracy, scalability, reliability and safety in line with responsible AI practices.

Project #2: Saleswhiz Enhancement V.2
Domain: Sales, Commercial AI Assistant Services
Duration: Aug 2025 – Feb 2026
Technology: Python, LLM, NLP, Vision AI, OCR, RAG, Agentic AI Pipeline, Embeddings, Vector Search, Hybrid Search, Cohere Re-Ranker, Content Chunking; SQL, Pinecone, Neo4j

Description: SalesWhiz is an AI-powered enterprise document intelligence and knowledge retrieval platform that extracts and processes business documents from SharePoint (PDF, Word, PowerPoint, Excel), converts them into structured AI-ready content, and enables intelligent semantic and hybrid search using embeddings, vector databases and Agentic AI pipelines to help sales and presales teams quickly access relevant insights.

Responsibilities:
- Implemented document extraction pipelines for PDF, DOCX, PPTX and Excel using Python-based processing libraries.
- Applied OCR and Vision AI models to process scanned documents and extract text from images.
- Converted extracted content into structured data with metadata enrichment using Python and JSON processing.
- Implemented content chunking and embedding generation using Azure OpenAI embedding models for semantic search.
- Indexed document embeddings in Pinecone vector database for high-speed vector retrieval.
- Developed Agentic AI retrieval pipelines using LLMs for intelligent query processing and response generation.
- Implemented hybrid search combining semantic vector search and lexical search techniques.
- Integrated Cohere Re-Ranker model to improve retrieval relevance and document ranking.

- Built intent detection and metadata extraction using LLM-based query understanding techniques.
- Performed testing, evaluation and optimization of retrieval pipelines to improve search accuracy and system performance.

DRDO (Defence Research and Development Organization)
Machine Learning Engineer (Intern) March 2025 – May 2025

Project: Spatio-Temporal Traffic Predictor
Domain: Transportation Analytics / Smart Traffic Management
Duration: March 2025 – May 2025
Technology: Python, ML, PyTorch, Deep Learning, STGCN, Graph Neural Networks (GNN), NumPy, Pandas

Description: Developed a traffic flow prediction system using STGCN to analyze spatial and temporal patterns in traffic data.

Responsibilities:
- Preprocessed traffic data from the PEMS dataset using Python.
- Implemented STGCN model using PyTorch for traffic prediction.
- Captured spatial and temporal dependencies in traffic networks.
- Trained and evaluated the deep learning model for prediction accuracy.
- Performed hyperparameter tuning (learning rate, batch size, dropout).
- Optimized model performance and validated prediction results.

CERTIFICATIONS

- Databricks Certified Generative AI Engineer Associate — Issued by Databricks, August 5, 2026 (Valid through August 5, 2028). Credential ID: 190654673. **View Certificate**